

A comprehensive review of the literature on continuous improvement approaches in food services management

Célia Monteiro^{a,b,c,*}, Ada Rocha^{a,c}, Vera Miguéis^d, Cláudia Afonso^{a,c,e}

^a Faculty of Nutrition and Food Sciences, University of Porto, Porto 4150-180, Portugal

^b Faculty of Sciences, University of Porto, Porto 4169-007, Portugal

^c GreenUPorto - Sustainable Agrifood Production Research Centre/Inov4Agro, Vairão 4485-646, Portugal

^d Faculdade de Engenharia da Universidade do Porto, INESC TEC, Rua Dr. Roberto Frias, Porto 4200-465, Portugal

^e Laboratory ITR, Laboratory for Integrative and Translational Research in Population Health, Institute of Public Health, University of Porto, Porto 4050-600, Portugal

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ABSTRACT

Continuous improvement (CI) have been recognised as one of the most effective ways to improve organisational performance. However, there is a lack of research on this topic from a food service perspective. Thus, the aim of this work is to explore the adoption of CI-focused methodologies in food services and to understand how they contribute to improving the performance of these services. Critical success factors and barriers to the implementation of CI are also analysed. This systematic review was conducted using the PRISMA methodology and a total of 43 studies were included in the analysis. This review shows that CI is effective in improving operations and performance, as well as increasing stakeholder satisfaction in the food service sector. Additionally, the review reveals that CI-focused tools are mainly used in problem identification, waste identification, planning, operations, and logistics. Human-related issues are the most frequently mentioned when it comes to the factors determining the success or failure of CI in food services.

1. Introduction

The food service (FS) industry involves a plethora of complex activities and typologies of services, such as commercial services (restaurants, catering services, and events), non-commercial and institutional services (business and education dining facilities), and different forms of care delivery (social, hospital, and others) (Martin-Rios et al., 2022). Despite the specificities of each sector, every FS includes activities related to food production management and meal service, from planning and preparation to delivery. Moreover, the different FS sectors share the responsibility of providing their customers with nutritionally and sensory-appropriate meals, as well as ensuring food safety and sustainability (Aytekin Sahin and Caferoglu, 2022; Martin-Rios et al., 2018).

Quality management is critical to the success of the FS industry, which aims to achieve a performance of excellence, increase customer satisfaction, reduce cycle times and costs, and eliminate errors and reworks (Gregoire, 2013). The recent COVID-19 pandemic and international conflicts have promoted global inflation, especially in the food, water, and energy sectors. This situation, together with the challenges posed by climate change, makes it even more imperative to optimise FS

management (Das, 2018; De Steur et al., 2016; Gładysz et al., 2020; Kahiluoto, 2020; McDermott et al., 2022).

Thus, it is important to improve the productivity and cost-effectiveness of FSs, and, at the same time, integrate sustainable practices throughout the meal production chain. Interventions must begin with a careful selection of food suppliers, the use of more efficient equipment, and a reduction of resources losses (Martin-Rios et al., 2022; Lefadola et al., 2018; Vizzoto et al., 2021), aligned with the Sustainable Development Goals (SDGs) for 2030 proposed by the United Nations. More specifically, these activities can contribute to goal number 2 (zero hunger) and goal number 12 (ensure sustainable consumption and production patterns) (United Nations, 2015).

Continuous improvement (CI) has long been regarded as one of the most effective ways to improve organisational performance and quality. This process can be useful to the FS industry for the purpose of improving process management. CI optimises the quality of products and services and create a culture of permanent improvement through a continuous search for better results. CI promotes the integration of quality into all aspects of a business or organisation, offering a set of tools and techniques designed to evaluate organisational performance,

* Corresponding author at: Faculty of Nutrition and Food Sciences, University of Porto, Porto 4150-180, Portugal.

E-mail address: up200801502@edu.fcna.up.pt (C. Monteiro).

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identify key points for improvement, and promote changes (Vinodh et al., 2021). In addition, when integrated into management approaches, CI has the potential to improve environmental sustainability and business performance (Das, 2018; De Steur et al., 2016; Gladysz et al., 2020; McDermott et al., 2022). Recent studies on a new combination of practices known as Lean Green corroborate this fact (Samant and Prakash, 2021; Skaymanska-Bralkowska and Malinowska, 2019).

Several studies have demonstrated the application of CI in different sectors (McDermott et al., 2022; Audretsch et al., 2011; Dora et al., 2016; Klee et al., 1994). In the FS context, studies focus on the applicability of specific CI-focused tools. No study has yet brought together all the approaches used in this sector to assist practitioners and researchers in exploring this topic in the FS sector.

Thus, in the present systematic literature review, we seek to explore the adoption of methodologies focused on CI in the FS industry and understand how they contribute to the optimisation of the services provided. We therefore propose the following research questions (RQs):

RQ1: What CI-focused methodologies, techniques, and tools are successfully used in the FS industry?

RQ2: What are the most used CI-focused tools in each type of FS?

RQ3: What are the main outcomes and improvements provided to each FS by CI-focused methodologies?

RQ4: Which factors are most often associated with the success or failure of a CI-focused methodologies implementation in the context of the FS industry?

The present paper is structured as follows. In Section 1, we introduced the research topic. In Section 2, we provide a brief description of the background knowledge on the main CI-focused methodologies. In Section 3, we describe the methodology used in the present systematic review. In Section 4, we reveal the results obtained, and, in Section 5, we discuss these results. Finally, in Section 6, we present the conclusions and provide suggestions for future work.

2. Background knowledge

Initially described by Deming as a philosophy made up of improvement initiatives that increase successes and reduce failures, CI has, over time, been recognised as one of the most effective ways of improving performance and quality in organisations (Juergensen, 2000).

Bhuiyan et al (Bhuiyan and Baghel, 2005). provided a broader definition of this concept, defining CI as a culture of sustained improvement aimed at eliminating waste in all of an organisation's systems and processes. In addition, CI requires everyone involved to work together, without necessarily involving large financial investments.

Over time, the concept of CI has progressively evolved in an organised and comprehensive way, becoming associated with methodologies based on a certain basic concept of quality or process improvement, with the aim of reducing waste, simplifying production flows, and improving quality (Bhuiyan and Baghel, 2005; García-Alcaraz and Oropesa-Vento, 2017).

The lean methodology, one of the best-known CI-focused methodologies, evolved from the Toyota Production System (TPS) in Japan (Bhuiyan and Baghel, 2005; Psomas, 2021), and leads to improvements by using a series of techniques and tools dedicated to identifying problems and eliminating everything that does not add value to a process. According to Garcia-Alcaraz et al., (García-Alcaraz and Oropesa-Vento, 2017) one of these tools is the *kaizen* system. For these authors, *kaizen* is the tool at the basis of the operational stability of the lean manufacturing pyramid, and only with this good foundation can other flow and quality tools be supported.

However, there seems to be no consensus in the literature regarding the definitions of CI, lean thinking, and *kaizen*, with many authors using these terms interchangeably (Chiarini et al., 2018; Singh and Singh, 2009).

Kaizen is the Japanese word for “improvement” and was defined by

Masaaki Imai (Imai, 1986) as “a means of continuous improvement in personal life, domestic life, social life, and professional life. In the workplace, *kaizen* means continuous improvement involving everyone — managers and workers alike”. Posteriorly, the same author stated that there are three different forms of *kaizen*: *kaizen* management, group *kaizen*, and individual *kaizen*. The first, *kaizen* management, refers to the organisational strategy behind management objectives and policy. Group *kaizen* focuses on teamwork to fulfil improvement cycles and solve daily problems. Finally, individual *kaizen* highlights the main role of operational workers in contributing to organisational improvements. In 1997, Berger (Berger, 1997) proposed three guiding principles for *kaizen*: 1, *process-oriented kaizen*, which is about improving processes in order to improve performance; 2, *kaizen focused on maintaining or improving standards*, i.e. improving or maintaining standards through continuous efforts by the organization; 3, *individual-oriented kaizen*, which focuses the role of improvement on the employee. Alongside this, Suárez-Barraza et al (Suárez-Barraza et al., 2011a). realised that in order to achieve results *Kaizen* needs a set of guiding principles that can provide a theoretical basis when implementing its theoretical concepts, i.e.: 1, *the basic elements* - simple things to implement to consolidate *kaizen*; 2, *maintaining and improving standards* – basic requirement of continuous improvement; 3, *the process-oriented approach* - focus efforts on improving the organisation's processes; 4, *the people-oriented approach* – substantial contribution of staff to improvement; 5, *CI in everyday work* – everyday improvement by solving problems at the workplace and by eliminating *muda*.

Brunet and New (Brunet and New, 2003), defined *kaizen* as “pervasive and continual activities, outside the contributor's explicit contractual roles, to identify and achieve outcomes he believes contribute to the organisational goals”, concluding that *kaizen* principals should be maintained despite the extensive practices, techniques and tools that are used, and give employees and their workforce the main role in the development of the company. In line with this, Aoki (Aoki, 2008) argues that the goal of *kaizen* is to achieve sustainable competitive advantage through the creation of innovative organizational capabilities. This study has identified three organizational capabilities, namely the ability to encourage employees to take initiative, the ability to facilitate interdepartmental staff communication and, finally, the ability to discipline workers.

Suárez-Barraza et al (Suárez-Barraza et al., 2011b). argue that the different ways of applying *kaizen* described in the literature converge into three perspectives of *kaizen*. One proposes *kaizen* as a management philosophy that encompasses Japanese management practices, techniques, and tools. Another perspective treats *kaizen* as a theoretical principle to underpin the application of improvement methodologies and techniques. The remaining one, proposes *kaizen* as a component of total quality management (TQM). From this perspective, *kaizen* is commonly referred to as “continuous improvement” and acts as a key element or value in the management system. Dahlgaard and Dahlgaard-Park (Dahlgaard and Dahlgaard-Park, 2006) defined TQM as “a company culture characterized by increased customer satisfaction through continuous improvement, in which all employees actively participate”. These authors conclude that the similarities between the objectives of Lean Thinking and TQM are no coincidence and that this is due to their roots in the evolution of Japanese quality, where Toyota was one of the pioneering companies (Dahlgaard and Dahlgaard-Park, 2006; Petersen, 1999).

Despite the ambiguity identified in these terms, in the present paper, we employ CI, lean thinking, and *kaizen* in line with the definitions of Bhuiyan et al (Bhuiyan and Baghel, 2005). and Garcia-Alcaraz et al (García-Alcaraz and Oropesa-Vento, 2017). mentioned above: CI as a cultural philosophy, lean thinking as a methodology and way of thinking, and *kaizen* as an improvement tool.

2.1. Lean thinking methodology

The lean thinking methodology has its roots in the TPS, created by Taiichi Ohno in early 1970s and developed by him in Japan (Bhuiyan and Baghel, 2005; Psomas, 2021). Labelled by Womack et al (Womack et al., 1990), and although initially applied only in the automotive industry, the lean thinking methodology has achieved enormous worldwide success and is currently applied in all areas of economic activity, from for-profit organisations to the public and governmental sectors and non-profit organisations (Shamah, 2013).

While lean manufacturing comes from manufacturing practices, the term lean production has emerged in the service sector, with a view to improving productivity and the quality of customer service (Bowen and Youngdahl, 1998; Levitt, 1972). Suárez-Barraza and Ramis-Pujol (Suárez-Barraza et al., 2011a), defined some fundamental aspects of lean service: strategic and operational objectives of the organization geared towards meeting customer needs efficiently and flexibly; application of the just in time (JIT) principle to stock control; management using value maps and flow diagrams to eliminate activities that do not add value; training staff in customer-oriented service and guiding customers themselves in building service quality; significant investment in employee involvement and participation. As an example, lean service was explored by Bowen and Youngdahl (Bowen and Youngdahl, 1998) in Taco Bell fast-food restaurants and compared the results with those obtained by McDonald's, which in 1960 chose to implement the mass-production line approach, centred on mass production and equipment technology. Despite rigid production and limited service, the company achieved profitability until, in 1980, it began to fall behind and had a large employee turnover. In contrast, Taco Bell chose to invest in people, understanding what customers valued, satisfying their needs, and betting on employee pay and performance and technology as support rather than replacement. Thus, the key to success was to let customer demand drive production, increasing the focus on quality customer.

The implementation of the lean management system, its concepts, and its spirit implies the development of an operational and cultural environment of CI in the long term (Psomas, 2021; Shamah, 2013; Bowen and Youngdahl, 1998; Levitt, 1972; Sorooshian and Fen, 2017; Clark et al., 2013; Pinto, 2014). Lean thinking has been applied worldwide as a concept and methodology of business leadership and management that aims to systematically eliminate waste and create value (Pinto, 2014). In this context, value can be defined as all activities that meet the expectations of stakeholders. In turn, waste, or *muda* in Japanese, is defined as anything that does not add value to a product or service and can refer to human or non-human activities and resources being used improperly and thus contributing to an increase in costs and cycle-time production and to the non-satisfaction of business stakeholders (Gregoire, 2013). Waste can be identified through the following techniques: the 3 MU — *muda* (waste), *mura* (variability), and *muri* (overflow); the 5 M+Q+S — men, machines, materials, management, method, quality, and safety; the flow of operations, referring to four actions — retention, transport, processing, and inspection; and the seven wastes, or 7 W — overproductions, waiting, transport, inappropriate processing, inventory, defects, and motion (Shamah, 2013; Hernade-wita, 2018). Moreover, a new category of waste has been mentioned: underutilisation of employee skills (García-Alcaraz and Oropesa-Vento, 2017).

One of the principles of the lean methodology is based on the pursuit of perfection. To achieve this, it is necessary to consider that the interests, needs, and expectations of the different stakeholders are constantly evolving (Pinto, 2014; Womack and Jones, 1996). To raise the standard of production quality and to meet or even exceed customer requirements, the lean methodology uses operational systems that maximise the added value of each element of an organisation. These systems, or tools, are fundamentally based on operational strategies, process design, quality management, and inventory management

(García-Alcaraz and Oropesa-Vento, 2017). Thus, the lean methodology encompasses a set of tools and techniques applied to different contexts and objectives. The following subsection summarises the most common ones.

2.1.1. Lean tools

With the evolution of the concept of CI and the lean methodology, several tools have been developed. In Table 1, we present the most common lean tools and techniques.

3. Systematic review methodology

3.1. Search strategy

We conducted the present systematic review according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement methodology (Page et al., 2021). A previous study where we researched and reviewed management approaches in the FS industry enabled a carefully selection on the databases and search terms to be used in the present systematic review. A search strategy was defined and performed the search in three databases: Scopus, Web of Science, and Business Source Ultimate (from EBSCO). The search was centred on two main groups of terms related to the FS industry and CI-focused methodologies, appropriately combined. The final query was: ("Food service" OR restaurant* OR catering OR canteen OR food-service) AND ("continuous improvement" OR "Lean service" OR "Lean management" OR "Lean tools" OR "Lean thinking" OR "quality management" OR "six sigma" OR kaizen OR "process improvement" OR "quality systems" OR (lean AND business) OR (lean AND (performance OR improvement OR effectiveness OR efficiency OR "cost reduction" OR optimiz*))). We searched for these terms in the title, keywords, and abstract fields.

The last search was in December 2022 and used the EndNote 20 reference management and bibliography software by Clarivate TM was used to assist us in the selection, screening, and management of the references. This systematic review was registered in the PROSPERO platform, with the identification code CRD42023393971.

3.2. Eligibility criteria

To ensure the relevance of the selected sources, we defined different eligibility criteria. First and foremost, we only accepted studies focusing on CI applied in the FS industry. Secondly, the CI-focused methodologies in question had to be either at the diagnostic stage or at an implementation stage. Additionally, we only assessed papers written in English, Portuguese, and Spanish were included. No filters regarding the publication date were applied, and all manuals, editorials, and commentaries were excluded.

3.3. Study selection

The screening and selection of the studies were carried out in three distinct phases to check their alignment with the objectives. We used the Rayyan software for systematic reviews (Qatar Computing Research Institute, Cambridge, USA/Doha, Qatar) (Ouzzani et al., 2016) to assist us with the selection and review process.

Initially, all the authors individually analysed each article by reading the titles and abstracts and decided for the inclusion or exclusion of the study. In case of doubt, the article would move on to the next selection phase. In the second phase, we read and analysed each of the selected articles in full to confirm their compliance with the eligibility criteria. In the third phase, all the authors held a meeting to discuss the exclusion of the articles.

Table 1
Most common lean tools and techniques.

Tool	Description	Examples of application
PDCA (plan-do-check-act) cycle	A facilitating element for the implementation of lean tools. It consists of a simple sequence, divided into four parts, which acts as a guide for CI by making changes and analysing situations.	(Faishal et al., 2021)
5S (<i>seiri, seiton, seiso, seiketsu, and shitsuke</i>)	A method for maintaining optimal conditions in the workplace, including a set of practices that seek to reduce waste and improve the performance of people and processes.	(Chapman, 2005; Leming-Lee et al., 2019)
VSM (value stream mapping)	A technique that enables the identification of the activities that create and deliver value to stakeholders, involving problem solving, information management, and physical transformation (from materials to final products).	(De Steur et al., 2016; Pinto, 2014; Hernadewita, 2018)
Spaghetti diagram	A diagram that results from recording the flows on the floor plan of the facilities.	(Samant and Prakash, 2021)
Visual management	A collection of techniques that encompasses visual communication and control of the work environment, facilitating the perception of the processes implemented and guiding teams to action (e. g. tables, graphs, and signposts).	(Kaizen Institute, 2016)
Root cause analysis	A systematic method for identifying and resolving the root causes of quality problems. Quality control tools include Pareto charts (or ABC analyses), scatter diagrams, check sheets, control charts, histograms, stratification, and cause-and-effect diagrams (known as fishbone or Ishikawa diagrams).	(Suárez-Barraza and Rodríguez-González, 2019; Ishikawa, 1976)
Pull system	A technique that uses the just-in-time (JIT) strategy, i.e. production of only the expected items, in the right quantity and at the right time. It is supported by the <i>kanban</i> system, which controls the flow of materials, people, and information on the shop floor (or <i>gemba</i>), and ensures the operation is controlled by the final customer.	(Shah and Ward, 2007)
DMAIC (define, measure, analyse, improve, and control) cycle	The main tool of the Six Sigma approach, which is used to stabilise and improve a process and sustain any improvements over the long term. In Six Sigma, the goal is to have a process that produces no more than 3.4 defects per	(Psomas, 2021; Imai, 1986)

Table 1 (continued)

Tool	Description	Examples of application
Total productive maintenance	million opportunities (DPMO), which is equivalent to a defect rate of 3.4 PPM. A strategic and comprehensive approach to maintenance that seeks to eliminate breakdowns and stoppages through the improvement of equipment efficiency, preventive maintenance by specialised technicians, autonomous maintenance by operators, training of all the people involved, and adequate design of the equipment for the facilities.	(Alsubaie and Yang, 2017)
QFD (quality function deployment)	A method for translating qualitative user requirements into quantitative parameters. It involves four phases: planning matrix (known as the "house of quality"), product development matrix, process/ component development matrix, and work instruction matrix.	(Huang et al., 2022)
VOC (voice of customer)	A tool used to listen to the consumer and seek customer-oriented solutions. It is normally used to start the application of a QFD.	(Fongsatitkul and Kainuma, 2021)
Hoshin kanri planning	A method used for the formal planning of policies and objectives to align and encourage employee involvement with an organisation's goals.	(Pinto, 2014)
Error proofing methods (mistake proofing, fool proofing, idiot proofing, and fail-safe)	Activities for identifying, preventing, eliminating, and correcting probable causes of errors or defects in processes and procedures to avoid affecting the quality of a product and the health and safety of operators. Also known as <i>poka-yoke</i> practices.	(Romero et al., 2022)
DFX (design for excellence)	A method that complements <i>poka-yoke</i> practices in product development, helping designers to develop products that are easier to manufacture, safer, and more economical in order to minimise production time and costs.	(Pinto, 2014)
Kaizen	A lean system that brings about small changes or incremental improvements in working methods or work processes. It reduces waste and consequently improves work performance, leading organisations into a spiral of increasing innovation.	(Suárez-Barraza et al., 2011b)

3.4. Data collection

We analysed the results of each selected study and included them in the present review. Relevant information was extracted and organised it in table format according to the following topics: general information — author, year, place of publication, and target FS sector; study characteristics — applied CI-focused methodologies and tools; study indicators — main aims and main outcomes (beneficial or not) of applying the CI; and main success factors, limitations, and barriers in the implementation of the CI-focused methodologies. These elements help formulate a comprehensive narrative that gives an overview of the characteristics of the selected studies.

4. Results

The search process found 826 results: 461 from Scopus, 209 from Web of Science, and 156 from Business Source Ultimate. After the selection procedure, we included 43 articles in the present review. A flow diagram detailing the search and selection process is shown in Fig. 1.

The selected studies were divided into two groups: those whose findings refer to the diagnosis phase (Appendix A) and those that implement CI-focused methodologies and present results (Appendix B). Diagnostic studies focus on understanding the initial state of processes, without implying an intervention, while implementation studies imply a CI-aimed intervention. In this section, we describe the main results obtained from this categorisation.

4.1. Study characteristics

The present systematic review considers 43 studies that address CI-focused methodologies in the FS industry. Of these, 16 studies only address the diagnosis phase, whereas 27 studies implement CI-focused methodologies. The first study considered in this review dates to 1994. Analysing the trend in the publication of the articles, most of the papers were published between 2016 and 2020 (Fig. 2). The studies included are distributed over four continents, with Asia having the most studies (39.5 %) (Abdelhadi, 2016; Abukhalifeh and Som, 2015; Bhasin et al., 2021; Chen and Chen, 2016; Chen et al., 2018; Feng et al., 2012; Hsiao et al., 2016; Indrawati et al., 2020; Bhandari, 2018; Krishnamoorthy and Kapadia, 1999; Pan et al., 2017; Purwojatmiko, 2020; Yong and Wilkinson, 2003; Pun and ho, 2001; Shimmura et al., 2019; Yan and Ma, 2015), followed by America (34.8 %) (Suárez-Barraza et al., 2011a; Bower, 1996; Causado-Rodriguez et al., 2019; Gutierrez-Yllu et al., 2022; Hopen, 2017; Ortiz et al., 2020; Kambli et al., 2020; Seifert, 2005; Suárez-Barraza et al., 2021; Vergara et al., 2019; Yamamoto et al., 2010; Apte and Reynolds, 1995; Bojanic and Rosen, 1994; Letort and Boudreaux, 1994; Ramdeen et al., 2007), Europe (23.3 %) (Gładysz et al., 2020; Krzemień and Wolniak, 2007; Ahmed et al., 2015; Engelund et al., 2009; Jaca et al., 2018; Morales-Contreras et al., 2020; Orynych et al., 2020; Strotmann et al., 2017; Teeling et al., 2019), and, finally, Oceania (2.3 %) (Cao et al., 2008). Fig. 4 shows the geographical distribution of the group of studies that address the diagnosis phase and the group of studies that implement CI-focused

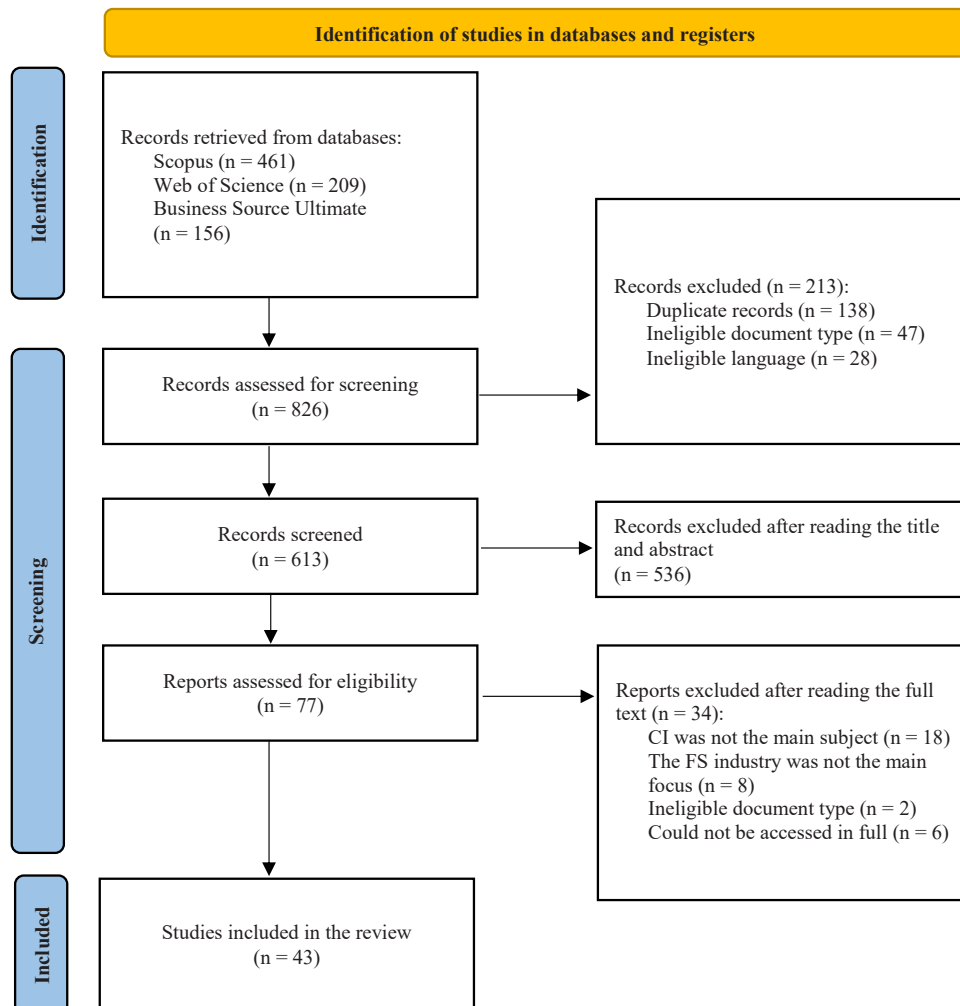


Fig. 1. PRISMA flow diagram showing the selection of articles dealing with CI approaches in the FS industry.

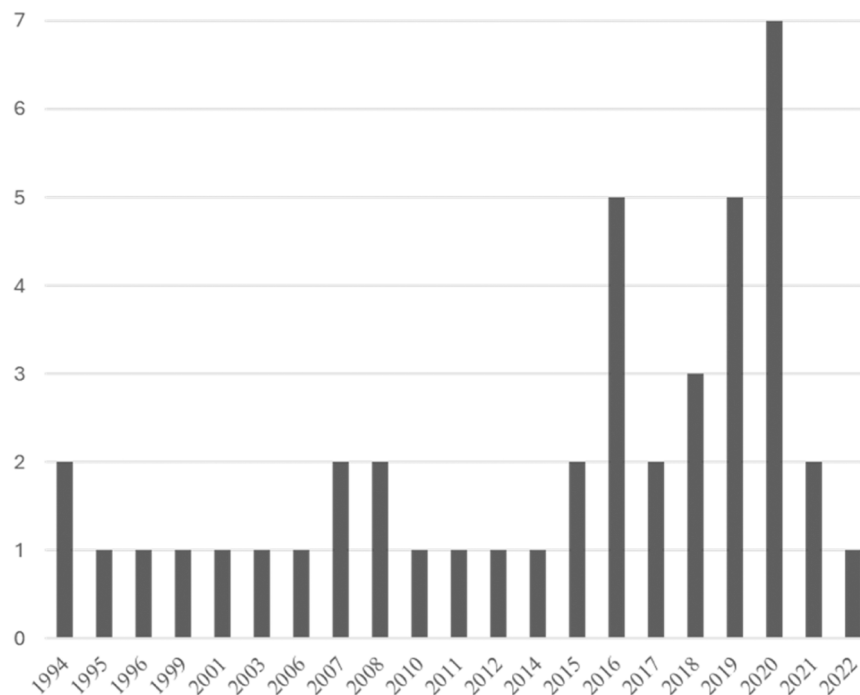


Fig. 2. Study publication trend analysis.

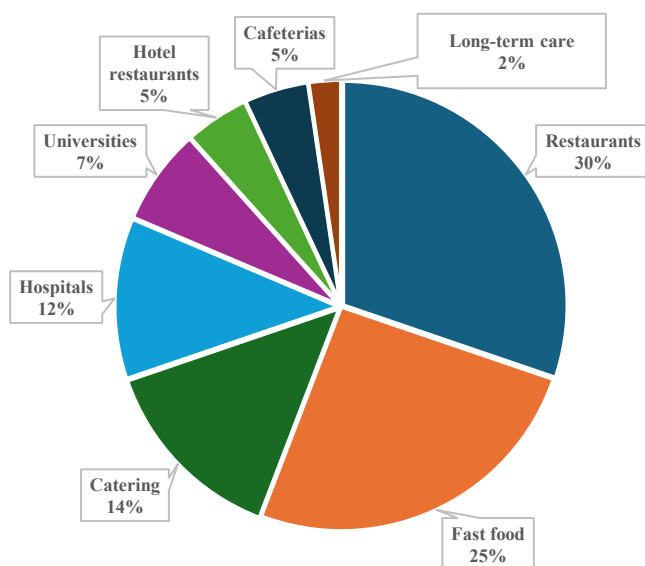


Fig. 3. General distribution of papers by food service sector.

methodologies. Regarding FS sectors, most of the studies take place in the commercial segment. The restaurant (30 %) (Suárez-Barraza et al., 2011a; Hsiao et al., 2016; Bhandari, 2018; Pun and ho, 2001; Shimmura et al., 2019; Yan and Ma, 2015; Causado-Rodriguez et al., 2019; Gutierrez-Yllu et al., 2022; Hopen, 2017; Ortiz et al., 2020; Vergara et al., 2019; Bojanic and Rosen, 1994; Cao et al., 2008) and fast-food (25 %) (Abdelhadi, 2016; Bhasin et al., 2021; Chen et al., 2018; Indrawati et al., 2020; Yong and Wilkinson, 2003; Suárez-Barraza et al., 2021; Apte and Reynolds, 1995; Krzemień and Wolniak, 2007; Morales-Contreras et al., 2020; Orynycz et al., 2020; Kanyan et al., 2016) sectors are the most common ones, followed by catering services (14 %) (Skaymanska-Bralkowska and Malinowska, 2019; Feng et al., 2012; Seifert, 2005; Jaca et al., 2018; Strotmann et al., 2017) and cafeterias (5 %) (Gładysz et al., 2020; Krishnamoorthy and Kapadia, 1999).

In the institutional segment, FSs in hospitals (12 %) (Yamamoto et al., 2010; Letort and Boudreaux, 1994; Ahmed et al., 2015; Engelund et al., 2009; Teeling et al., 2019) and FSs in universities (7 %) (Chen and Chen, 2016; Pan et al., 2017; Kambli et al., 2020) are the types of FS that implement CI-focused methodologies the most. We also found relevant papers in other, less-studied sectors: long-term care facilities (Bower, 1996), hotel restaurants (Abukhalifeh and Som, 2015; Ramdeen et al., 2007), inns (Gładysz et al., 2020), railways (Gładysz et al., 2020), and food stalls (Purwojatmiko, 2020) (Fig. 3).

Analysing the diagnostic and implementation studies separately (Fig. 5), we can conclude that in both types, the most studied sectors are the restaurant sector, with six diagnostic studies (Bhandari, 2018; Pun and ho, 2001; Causado-Rodriguez et al., 2019; Ortiz et al., 2020; Bojanic and Rosen, 1994; Cao et al., 2008) and seven implementation studies (Suárez-Barraza et al., 2011a; Hsiao et al., 2016; Shimmura et al., 2019; Yan and Ma, 2015; Gutierrez-Yllu et al., 2022; Hopen, 2017; Vergara et al., 2019), and the fast-food sector, with six diagnostic studies (Abdelhadi, 2016; Chen et al., 2018; Yong and Wilkinson, 2003; Krzemień and Wolniak, 2007; Morales-Contreras et al., 2020; Kanyan et al., 2016) and five implementation studies (Bhasin et al., 2021; Indrawati et al., 2020; Suárez-Barraza et al., 2021; Apte and Reynolds, 1995; Orynycz et al., 2020). On the other hand, FSs in hospitals and FSs in university canteens are more common in implementation studies, with four (Yamamoto et al., 2010; Letort and Boudreaux, 1994; Engelund et al., 2009; Teeling et al., 2019) and three studies (Chen and Chen, 2016; Pan et al., 2017; Kambli et al., 2020), respectively, than in diagnostic studies, with one study (Ahmed et al., 2015) and zero studies, respectively. There is only one implementation study in the social sector (Bower, 1996).

4.2. Methodologies and main outcomes

The studies included in this review reveal references to lean, six sigma, lean six sigma and TQM methodologies (Appendices A and B). The frequencies of their application in FS sector are the following: lean methodology, mentioned in 11 studies - three in restaurants (Bhandari, 2018; Gutierrez-Yllu et al., 2022; Vergara et al., 2019), three in fast-food (Abdelhadi, 2016; Morales-Contreras et al., 2020; Orynycz

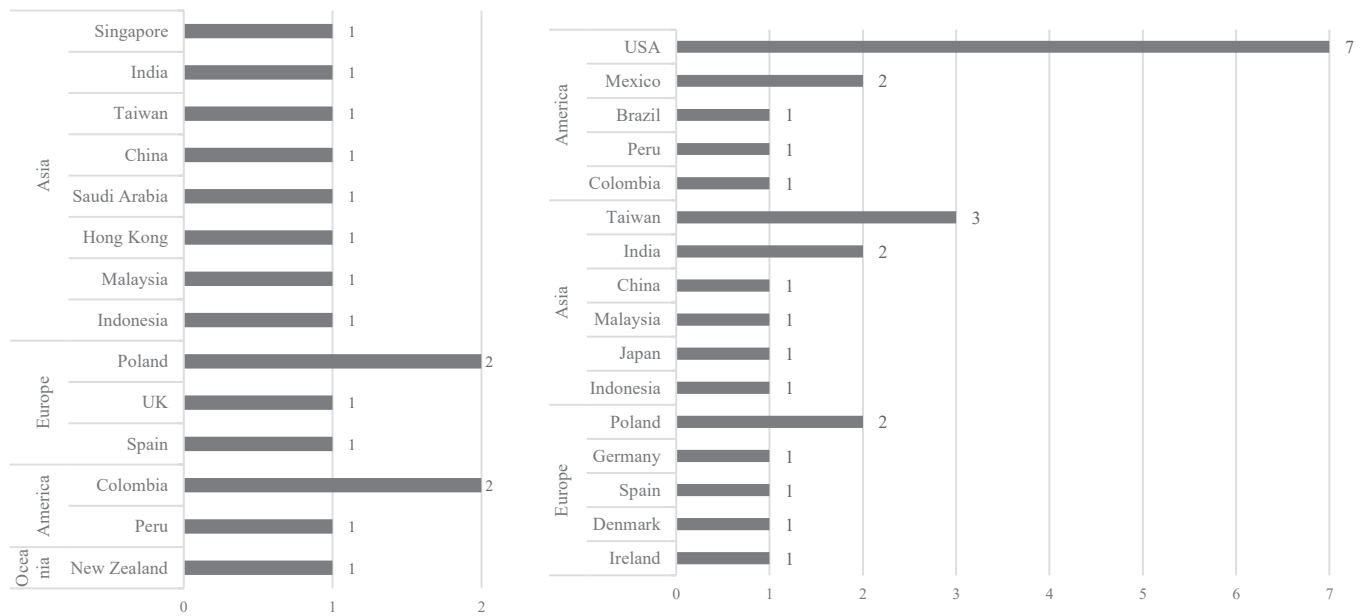


Fig. 4. Geographical distribution of diagnostic (left) and implementation (right) studies on CI, n.

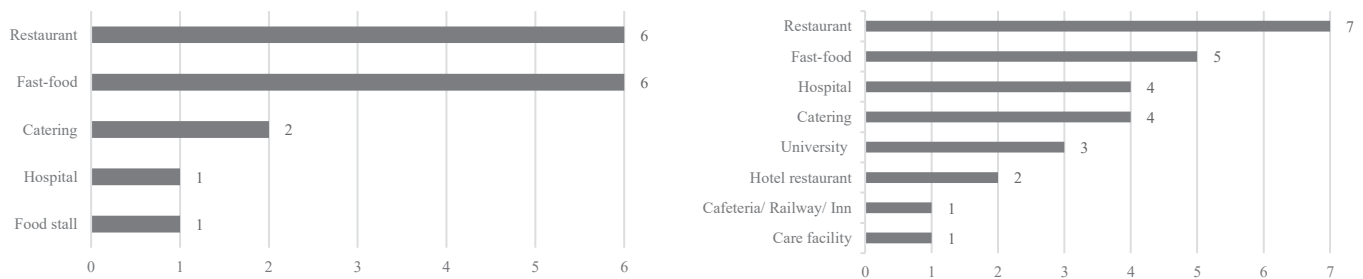


Fig. 5. Distribution of diagnostic (left) and implementation (right) studies on CI by FS sector, n.

et al., 2020), two in hospital facilities (Suárez-Barraza et al., 2021; Engelund et al., 2009), one in university facilities (Pan et al., 2017), one in catering sectors (Skaymanska-Bralkowska and Malinowska, 2019) and one in cafeterias (Gładysz et al., 2020); six sigma methodology, mentioned in six studies - two in restaurants (Hsiao et al., 2016; Ortiz et al., 2020), two in university facilities (Chen and Chen, 2016; Kambli et al., 2020), two in catering sector (Feng et al., 2012; Seifert, 2005), one in fast-food (Bhasin et al., 2021); lean six sigma methodology mentioned in four studies: two in hospital facilities (Yamamoto et al., 2010; Teeling et al., 2019), one in restaurant (Hopen, 2017), and one in fast-food (Indrawati et al., 2020); and TQM, mentioned in five studies: two in fast-food (Yong and Wilkinson, 2003; Kanyan et al., 2016), one in restaurant (Bojanic and Rosen, 1994), one in catering sector (Strotmann et al., 2017), one in cafeteria (Purwojatmiko, 2020), and one in hotel restaurant (Abukhalifeh and Som, 2015). The other studies included in this review did not clearly specify the methodology used, but instead applied tools.

Regarding the diagnostic studies, few apply CI-focused tools. Regardless, the most common are VOC (two studies) and VSM (two studies). On the other hand, the studies that implement CI-focused methodologies apply a wide range of lean tools, the most common of which are the DMAIC cycle (ten studies), the Ishikawa diagram (nine studies), the Pareto chart (eight studies), and *kaizen* (seven studies). Fig. 6 shows in more detail the frequency of the methodologies applied in the diagnostic studies and in the studies that implement CI.

Table 2 presents a summary of the CI-focused tools applied in the two types of studies considered in this review, grouped according to their

functionality and by FS sector. According to mentioned above, some general tools act as an umbrella for other specific tools (e.g. *kaizen* and 5S). However, the general and specific tools mentioned in the studies analysed are considered equally when classifying functionality. The CI initiatives most commonly applied in the FS industry relate to problem identification and resolution (Ishikawa diagram (Skaymanska-Bralkowska and Malinowska, 2019; Bhasin et al., 2021; Chen and Chen, 2016; Bower, 1996; Hopen, 2017; Ortiz et al., 2020; Suárez-Barraza et al., 2021; Yamamoto et al., 2010; Apte and Reynolds, 1995; Letort and Boudreaux, 1994), Pareto chart (Suárez-Barraza et al., 2011a; Bhasin et al., 2021; Chen and Chen, 2016; Hsiao et al., 2016; Gutierrez-Yllu et al., 2022; Hopen, 2017; Ortiz et al., 2020; Seifert, 2005; Apte and Reynolds, 1995; Letort and Boudreaux, 1994), flowchart (Bhasin et al., 2021; Hsiao et al., 2016; Seifert, 2005; Yamamoto et al., 2010; Apte and Reynolds, 1995; Letort and Boudreaux, 1994), and 5Why (Gładysz et al., 2020); Skaymanska-Bralkowska and Malinowska, 2019; Teeling et al., 2019), followed by planning, operations, and logistics characterisation (PDCA cycle (Gładysz et al., 2020; Pan et al., 2017; Suárez-Barraza et al., 2021; Apte and Reynolds, 1995; Letort and Boudreaux, 1994; Orynycz et al., 2020; Strotmann et al., 2017; Teeling et al., 2019), *kaizen* (Gładysz et al., 2020; Skaymanska-Bralkowska and Malinowska, 2019; Suárez-Barraza et al., 2011a, 2021; Engelund et al., 2009; Orynycz et al., 2020; Strotmann et al., 2017), visual management (Gładysz et al., 2020; Causado-Rodriguez et al., 2019; Engelund et al., 2009), *kanban* (Gładysz et al., 2020; Skaymanska-Bralkowska and Malinowska, 2019), JIT and pull system (Skaymanska-Bralkowska and Malinowska, 2019; Apte and Reynolds, 1995), *poka-yoke* (Gutierrez-Yllu

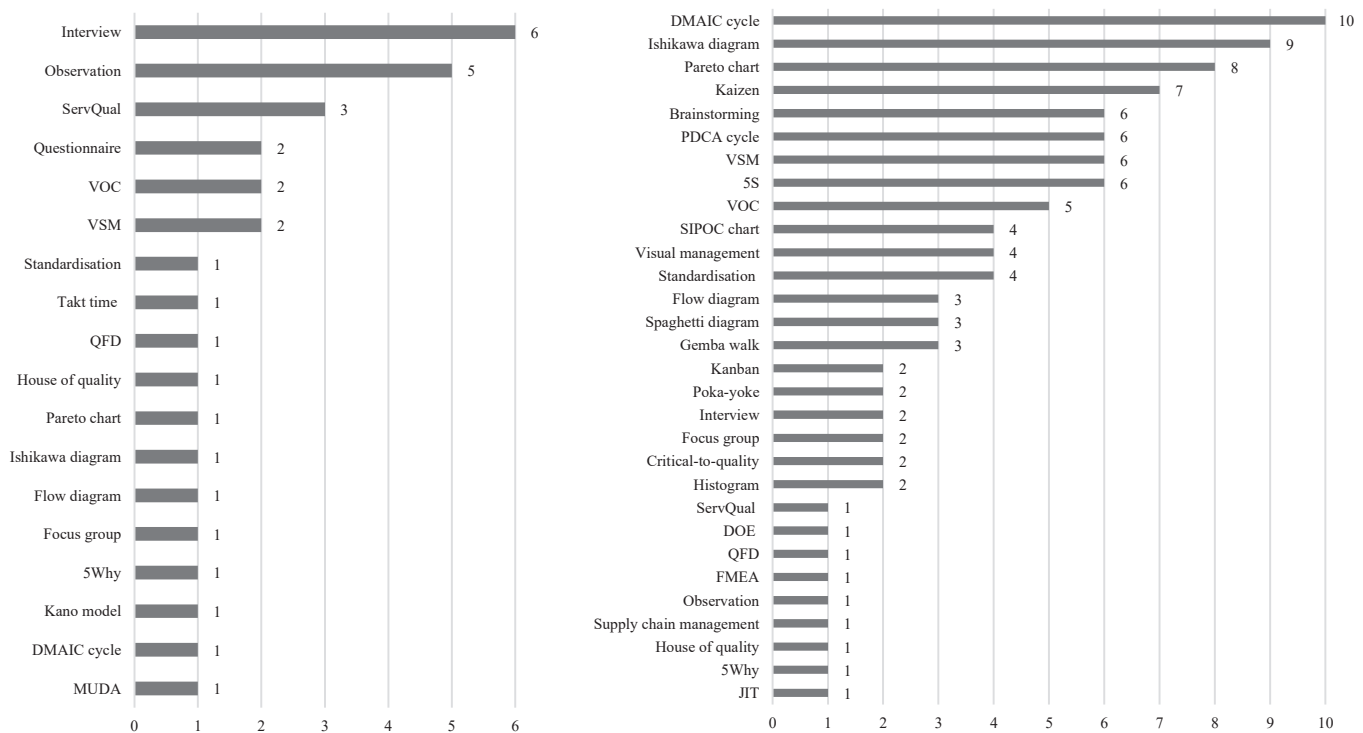


Fig. 6. Tools used in the diagnostic (left) and implementation (right) studies, n. 5S: seiri, seiton, seiso, seiketsu, and shitsuke; VSM: value stream mapping; PDCA: plan-do-check-act cycle; JIT: just-in-time strategy; QFD: quality function deployment; VOC: voice of customer; FMEA: failure mode and effects analysis; SIPOC: suppliers, inputs, process, outputs, and customers; DMAIC: define, measure, analyse, design, and control; DOE: design of experiments.

et al., 2022; Ramdeen et al., 2007), and takt time (Yong and Wilkinson, 2003)) and waste elimination (VSM⁷, 15, 62, 75, 79, 86, 87, 90–82, 5S (Skaymanska-Bralkowska and Malinowska, 2019; Suárez-Barraza et al., 2011a; Indrawati et al., 2020; Bhandari, 2018; Engelund et al., 2009; Orynych et al., 2020), and muda (Bhandari, 2018); Morales-Contreras et al., 2020). Six sigma tools (Bhasin et al., 2021; Chen and Chen, 2016; Chen et al., 2018; Hsiao et al., 2016; Indrawati et al., 2020; Krishnamoorthy and Kapadia, 1999; Ortiz et al., 2020; Kambli et al., 2020; Seifert, 2005; Yamamoto et al., 2010; Orynych et al., 2020; Teeling et al., 2019) (DMAIC cycle, SIPOC diagram, DOE, and the Kano model) and typical value creation tools (VOC (Bhasin et al., 2021; Chen and Chen, 2016; Bhandari, 2018; Hopen, 2017; Seifert, 2005), standardisation (Gładysz et al., 2020; Skaymanska-Bralkowska and Malinowska, 2019; Suárez-Barraza et al., 2011a; Feng et al., 2012; Gutierrez-Yllu et al., 2022), house of quality and QFD (Chen et al., 2018; Yan and Ma, 2015), DFX (Seifert, 2005), and FMEA (Seifert, 2005) have also been implemented, although less frequently. From an individual FS perspective, problem identification and resolution tools are the most common tools in all sectors with the exception of catering services and cafeterias, where the most commonly applied methodologies relate to planning, operations, and logistics (Gładysz et al., 2020; Skaymanska-Bralkowska and Malinowska, 2019; Feng et al., 2012; Krishnamoorthy and Kapadia, 1999; Purwojatmiko, 2020; Seifert, 2005; Jaca et al., 2018; Strotmann et al., 2017).

The main outcomes were divided into five groups according to their typology. They are, in descending order of citation frequency: 1) operational and performance improvements (73 % of studies), which includes results related to productivity improvement, effectiveness and efficiency operations, production time improvement (lead and cycle time), number of operations, and rework reductions (Gładysz et al., 2020; Suárez-Barraza et al., 2011a; Abukhalifeh and Som, 2015; Bhasin et al., 2021; Chen and Chen, 2016; Chen et al., 2018; Hsiao et al., 2016; Indrawati et al., 2020; Bhandari, 2018; Pan et al., 2017; Yong and Wilkinson, 2003; Yan and Ma, 2015; Bower, 1996; Causado-Rodriguez et al., 2019; Gutierrez-Yllu et al., 2022; Hopen, 2017; Kambli et al.,

2020; Seifert, 2005; Suárez-Barraza et al., 2021; Vergara et al., 2019; Yamamoto et al., 2010; Apte and Reynolds, 1995; Letort and Boudreaux, 1994; Engelund et al., 2009; Jaca et al., 2018; Orynych et al., 2020; Teeling et al., 2019); 2) stakeholder satisfaction improvements (59 % of studies), which includes results related to customer satisfaction, reductions in waiting times and complaints, improvements in meal quality, customer clinical and nutritional status, and employee satisfaction and empowerment (Gładysz et al., 2020; Suárez-Barraza et al., 2011a; Abukhalifeh and Som, 2015; Chen and Chen, 2016; Hsiao et al., 2016; Krishnamoorthy and Kapadia, 1999; Pan et al., 2017; Yong and Wilkinson, 2003; Yan and Ma, 2015; Bower, 1996; Causado-Rodriguez et al., 2019; Hopen, 2017; Ortiz et al., 2020; Kambli et al., 2020; Seifert, 2005; Vergara et al., 2019; Yamamoto et al., 2010; Apte and Reynolds, 1995; Letort and Boudreaux, 1994; Engelund et al., 2009; Jaca et al., 2018; Teeling et al., 2019); 3) financial improvements (24 % of studies), which includes cost savings, inventory reduction, and profitability (Hsiao et al., 2016; Bhandari, 2018; Krishnamoorthy and Kapadia, 1999; Vergara et al., 2019; Apte and Reynolds, 1995; Ramdeen et al., 2007; Strotmann et al., 2017; Teeling et al., 2019); 4) improvements in environmental sustainability factors (14 % studies), which includes improvements related to food waste reduction, natural resource savings, and CO₂ emissions reduction (Gładysz et al., 2020; Skaymanska-Bralkowska and Malinowska, 2019; Engelund et al., 2009; Orynych et al., 2020; Strotmann et al., 2017); and 5) competitiveness improvement (25 % of studies), with results related to market value changes (Orynych et al., 2020; Strotmann et al., 2017).

4.3. Factors associated with the success or failure of CI implementations

To identify the factors that contribute to the success or failure of a CI initiative's implementation in the FS industry, we prepared a list with all the critical success factors and all the barriers mentioned in the studied papers. We found 173 mentions and grouped them by approximation of characteristics into the following categories and subcategories: *methods* — 24 mentions (methods and tools, methodology integration, and

Table 2

Summary of the main CI-focused tools identified in the studies under analysis and main results.

Food service sector	Author and reference	Waste elimination			Problem identification and resolution				Planning, operations, and logistics							Value creation					Six Sigma				Outcomes
		5S	Muda	VSM	5Why	Pareto	Ishikawa	Flowchart	PDCA	Kanban	JIT/ Pull	Kaizen	Visual board	Poka-yoke	Takt time	Stand.	DFX	QFD HOQ	VOC	FMEA	SIPOC	DMAIC	DOE	Kano	
Restaurants	Gutierrez-Yllu et al., 2022 (Gutierrez-Yllu et al., 2022)					X								X		X									a, c
	Ortiz Porras et al., 2020 (Ortiz et al., 2020)			X		X	X	X														X			b
	Vergara et al., 2019 (Vergara et al., 2019)			X																					a, b
	Causado-Rodriguez et al., 2019 (Causado-Rodriguez et al., 2019)											X													a, b
	K. S & Bhandari, 2018 (Bhandari, 2018)	X	X																X						a, c
	Hopen, 2017 (Hopen, 2017)					X	X												X			X			a, b
	Hsiao et al., 2016 (Hsiao et al., 2016)					X		X													X	X			a, b, c
	Yan & Ma, 2015 (Yan and Ma, 2015)																	X							a, b
	Suárez-Barraza et al., 2011 (Suárez-Barraza et al., 2011a)	X										X				X									a, b, c
	Suárez-Barraza et al., 2021 (Suárez-Barraza et al., 2021)					X	X		X			X													a
Fast-food	Bhasin et al., 2021 (Bhasin et al., 2021)						X	X											X		X	X			a
	Morales-Contreras et al., 2020 (Morales-Contreras et al., 2020)		X																						NA
	Orynycz et al., 2020 (Orynycz et al., 2020)	X		X					X			X										X			a, d, e
	Indrawati et al., 2020 (Indrawati et al., 2020)	X		X																		X			a
	Chen et al., 2018 (Chen et al., 2018)																	X					X		NA
	Abdelhadi, 2016 (Abdelhadi, 2016)														X										NA
	Apte & Reynolds, 1995 (Apte and Reynolds, 1995)					X	X	X	X		X														a, b, c
	Skaymanska-Bralkowska & Malinowska, 2019 (Skaymanska-Bralkowska and Malinowska, 2019)	X		X	X		X			X	X	X				X									d
	Jaca et al., 2018 (Jaca et al., 2018)											X													a, b
	Strotmann et al., 2017 (Strotmann et al., 2017)			X					X																c, d, e
Catering	Feng et al., 2012 (Feng et al., 2012)															X									a
	Seifert, 2005 (Seifert, 2005)					X		X									X		X	X		X			a, b
	Ramdeen et al., 2007 (Ramdeen et al., 2007)												X												c
	Teeling et al., 2019 (Teeling et al., 2019)			X	X				X												X	X			a, b, c
Hotel restaurant																									
Hospitals and																									

(continued on next page)

Table 2 (continued)

Food service sector	Author and reference	Waste elimination			Problem identification and resolution				Planning, operations, and logistics							Value creation					Six Sigma					Outcomes
		5S	Muda	VSM	5Why	Pareto	Ishikawa	Flowchart	PDCA	Kanban	JIT/ Pull	Kaizen	Visual board	Poka- yoke	Takt time	Stand.	DFX	QFD HOQ	VOC	FMEA	SIPOC	DMAIC	DOE	Kano		
long-term care	Ahmed et al., 2015 (Ahmed et al., 2015)			X																					NA	
	Yamamoto et al., 2010 ⁸⁰						X	X													X	X			a, b	
	Engelund et al., 2009 (Engelund et al., 2009)	X		X							X	X													a, b, d	
	Bower, 1996 (Bower, 1996)						X																		a, b	
	Letort & Boudreaux, 1994 (Letort and Boudreaux, 1994)					X	X	X	X																a, b	
University	Kambli et al., 2020 (Kambli et al., 2020)																					X			a, b	
	Pan et al., 2020 (Pan et al., 2017)								X																a, b	
	Chen & Chen, 2016 (Chen and Chen, 2016)					X	X											X				X			a, b	
Cafeterias and others	Gładysz et al., 2020 (Gładysz et al., 2020)			X	X				X	X		X	X			X									a, b, d	
	Krishnamoorthy & Kapadia, 1999 ⁶⁴																						X		b, c	

Main outcomes: a - Operational and performance improvements; b - Stakeholder satisfaction improvements; c - Financial improvements; d - Sustainability factors improvement; e - Competitiveness improvement; NA - Not applicable; 5S: *seiri*, *seiton*, *seiso*, *seiketsu*, and *shitsuke*; VSM: value stream mapping; PDCA: plan-do-act-check cycle; JIT: just-in-time strategy; Pull: pull system; Stand.: standardisation; DFX: design for excellence; QFD: quality function deployment; HOQ: house of quality; VOC: voice of customer; FMEA: failure mode and effects analysis; SIPOC: suppliers, inputs, process, outputs, and customers; DMAIC: define, measure, analyse, design, and control; DOE: design of experiments.

knowledge of methods); *organisation* — 41 mentions (system characteristics, communication system, and workspace); *measures* — 17 mentions (indicators, available data, and consumer perspective); *human issues* — 45 mentions (employee involvement, training and skills, and motivation and attitudes); *resources* — 22 mentions (time and resources, and innovation and technology); and *environment* — 24 mentions (top management commitment, organisational culture, and rewards system). The Ishikawa diagram in Fig. 6 illustrates the distribution of the mentioned factors.

The most cited category of factors is *human issues*, where training and skills seem to be the most decisive factors. Knowledge background, systematic training, and employee involvement and empowerment are also cited as success factors (Gładysz et al., 2020; Skaymanska-Bralkowska and Malinowska, 2019; Suárez-Barraza et al., 2011a; Abdelhadi, 2016; Abukhalifeh and Som, 2015; Chen and Chen, 2016; Chen et al., 2018; Krishnamoorthy and Kapadia, 1999; Purwojatmiko, 2020; Yong and Wilkinson, 2003; Pun and ho, 2001; Shimmura et al., 2019; Ortiz et al., 2020; Suárez-Barraza et al., 2021; Yamamoto et al., 2010; Apte and Reynolds, 1995; Ramdeen et al., 2007; Jaca et al., 2018; Teeling et al., 2019). In turn, these topics can directly inspire the motivation, commitment, performance, and attitudes that can drive CI to synergistic levels (Suárez-Barraza et al., 2011a; Abukhalifeh and Som, 2015; Bhasin et al., 2021; Apte and Reynolds, 1995; Teeling et al., 2019). On the other hand, a lack of knowledge and training and a lack of understanding of CI-focused methodologies (Skaymanska-Bralkowska and Malinowska, 2019; Hsiao et al., 2016; Purwojatmiko, 2020; Ortiz et al., 2020; Ahmed et al., 2015; Orynych et al., 2020) lead to resistance to change and to demotivation, contributing negatively to CI (Suárez-Barraza et al., 2011a; Hsiao et al., 2016; Yong and Wilkinson, 2003; Bower, 1996; Ortiz et al., 2020; Letort and Boudreaux, 1994; Ahmed et al., 2015). Fig. 7

The second most cited category of factors is *organisation*. This category relates to the specificities of the services, facilities, and companies' work dynamics. Some studies specifically mention that redesigning the layouts and properly organising the services were critical factors for success (Gładysz et al., 2020; Suárez-Barraza et al., 2011a; Vergara et al., 2019). Other studies refer to small infrastructures, inadequate layouts, and disorganised work areas as barriers to CI (Abdelhadi, 2016; Indrawati et al., 2020; Bhandari, 2018; Purwojatmiko, 2020; Bower, 1996; Seifert, 2005; Ahmed et al., 2015; Morales-Contreras et al., 2020). Stakeholder inter- and intra-communication are also frequently mentioned. Internal communication between departments and between operational employees and the middle or top management (Yong and

Wilkinson, 2003; Pun and ho, 2001; Suárez-Barraza et al., 2021; Ramdeen et al., 2007) was considered a critical success factor most of the time, followed by having a good relationship with suppliers and other partners and company competitiveness (Bhandari, 2018; Pun and ho, 2001; Yamamoto et al., 2010; Ahmed et al., 2015; Strotmann et al., 2017).

Additionally, FSs are complex, with specificities that require additional care, usually associated with the diversity and perishability of the raw materials and the perishability of meals. Thus, these services are subject to food safety criteria and must comply with legal requirements. This topic was almost always considered a barrier to CI, due to the complexity and characteristics of the raw materials and processes, as they imply special routines and unexpected movements (Chen and Chen, 2016; Indrawati et al., 2020; Yong and Wilkinson, 2003; Ahmed et al., 2015; Englund et al., 2009).

The third most represented category is *methods*, which includes factors related to the suitability of CI-focused methodologies in relation to FS sectors and implementation objectives. Several authors state that the integration of various CI-focused methods among themselves or with other types of methodologies allows better results (Suárez-Barraza et al., 2011a; Abdelhadi, 2016; Indrawati et al., 2020; Causado-Rodriguez et al., 2019; Gutierrez-Yllu et al., 2022; Kambli et al., 2020; Seifert, 2005; Suárez-Barraza et al., 2021; Vergara et al., 2019; Englund et al., 2009; Strotmann et al., 2017) as well as the creation of innovative and more effective methodologies (Indrawati et al., 2020). Contrariwise, insufficient knowledge of CI-focused methodologies (Suárez-Barraza et al., 2011a; Purwojatmiko, 2020; Jaca et al., 2018; Orynych et al., 2020; Strotmann et al., 2017) or their inappropriate use (Gładysz et al., 2020; Skaymanska-Bralkowska and Malinowska, 2019; Bhandari, 2018) can be serious barriers to the success of projects. For example, isolated improvement efforts in a certain operation or certain subsection of an FS (island approach) can condition the potential benefit of the methodology (Gładysz et al., 2020; Suárez-Barraza et al., 2011a). Thus, CI should be transversal to the whole organisation.

The *environment* category is at the same level as the *methods* category. Several authors identify top management involvement and commitment as determining factors. Having an assertive, enthusiastic, and experienced leadership sends a positive message to work teams, motivating them and generating a sense of belonging that leads to greater participation in the pursuit of company goals (Hsiao et al., 2016; Purwojatmiko, 2020; Yong and Wilkinson, 2003; Pun and ho, 2001; Teeling et al., 2019). On the other hand if managers fail to recognise the value of CI may trigger the opposite feeling and consequent failure of the project

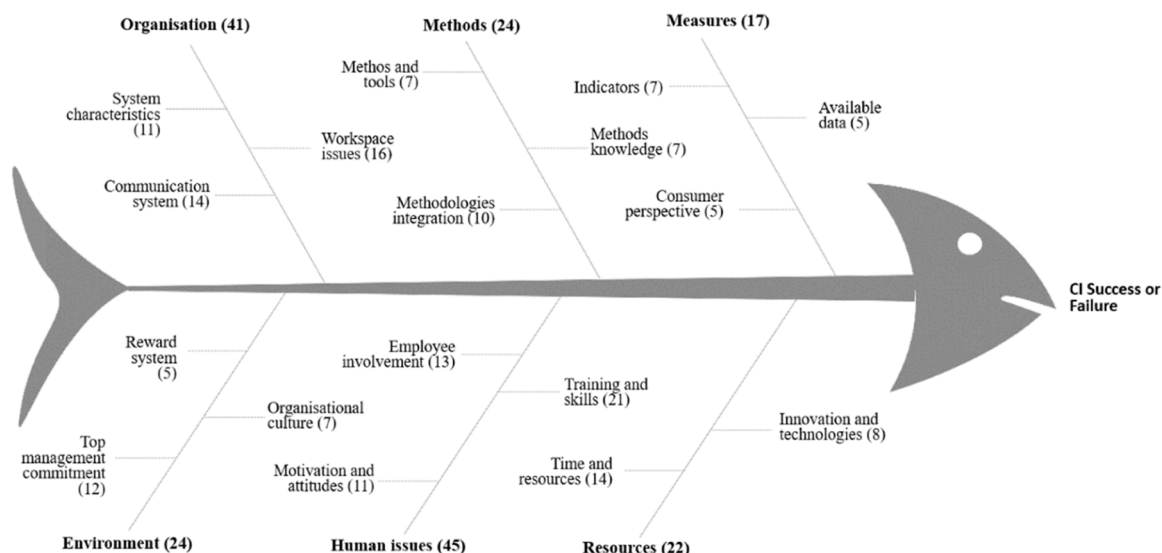


Fig. 7. Ishikawa diagram with the factors that contribute to or hinder the effectiveness of CI initiatives in the FS industry, n.

(Suárez-Barraza et al., 2011a; Pan et al., 2017; Causado-Rodriguez et al., 2019; Strotmann et al., 2017). A company's organisational culture is the basis for the implementation of the CI philosophy, according to some studies (Chen and Chen, 2016; Yong and Wilkinson, 2003; Apte and Reynolds, 1995; Ahmed et al., 2015). Additionally, competitive salaries and a system of incentives and benefits are considered important factors for success (Yong and Wilkinson, 2003; Apte and Reynolds, 1995; Krzemiński and Wolniak, 2007).

The *resources* category encompasses two types of parameters. The first refers to the time and resources required to implement and maintain CI-focused methodologies, which are considered a major obstacle to the success of lean initiatives. Several studies refer to the costs and the inability to allocate human resources specifically for the implementation of CI-focused methodologies as serious constraints on the success of operations (Gładysz et al., 2020; Chen and Chen, 2016; Bhandari, 2018; Yong and Wilkinson, 2003; Bower, 1996; Causado-Rodriguez et al., 2019; Gutierrez-Yllu et al., 2022; Kambli et al., 2020; Yamamoto et al., 2010; Strotmann et al., 2017; Teeling et al., 2019). Additionally, innovation and the adoption of new technologies are considered in some studies as promoters of success (Chen and Chen, 2016; Chen et al., 2018; Bhandari, 2018; Purwojatmiko, 2020; Yong and Wilkinson, 2003; Pun and Ho, 2001; Ortiz et al., 2020). In fact, innovation is an inherent component of CI, so it should be a requirement for the implementation of CI-focused methodologies.

Finally, the last category considered in the analysis is *measures*, which includes factors related to the evaluation of the effectiveness of the applied methodologies. Most of the analysed studies refer to this category as a barrier to CI implementation. The inadequate definition or application of key performance indicators and insufficient or excessive data are factors prone to the subjectivity of methodology effectiveness (Gładysz et al., 2020; Abdelhadi, 2016; Hsiao et al., 2016; Gutierrez-Yllu et al., 2022; Ortiz et al., 2020; Morales-Contreras et al., 2020). In addition, some studies mention customer perspective as a barrier to CI implementation, in the sense that personal and individual customer preferences result in subjective assessments. Customer's absence or reluctance to express an opinion or make a complaint is also considered a barrier to CI implementation (Bhasin et al., 2021; Hsiao et al., 2016; Bojanic and Rosen, 1994).

5. Discussion

The 43 studies considered in this analysis demonstrate the scarce research that exists in this sector. Nevertheless, there is a visible growing trend in the number of studies on CI applied to the FS industry. The first relevant studies about CI in the FS industry were published in 1994, but the growing trend is visible mainly after 2016, with a peak in 2020, when seven relevant studies were published.

The results obtained in the present research show that most of the studies took place in the commercial segment, namely in traditional restaurants and fast-food restaurants. Although less frequent, there are also some studies in the institutional segment, namely in hospital and university canteens. In this segment, care homes are the least studied locations. Additionally, there is a small number of relevant studies about FSs in food stalls, railways, and inns. It is interesting to note that despite the reduced number of studies focusing on CI-focused methodologies in the FS context, all FS segments have been explored in these studies. This confirms what we expected, i.e. the great potential for the adoption of CI-focused methodologies in any type of FS.

Diagnostic studies are essential to identify starting points for the implementation of CI. They provide examples of already implemented practices and common causes of waste along the value chain. In addition, they showcase quality attributes from the customers' point of view. Even though all of these studies envisage the subsequent implementation of CI, in some cases, some improvements are already obtained at this diagnostic stage. In addition, diagnostic studies are very valuable since their results allow us to determine the main intervention factors for

improvement. In relation to studies that implement CI-focused tools, the main motivations are the search for better performance and improved consumer satisfaction. These results differ from those found by Costa et al (Costa et al., 2018). These authors state that, in the food industry, the greatest motivation behind implementing CI-focused tools relates to cost reduction.

Our analysis shows that a wide range of CI-focused tools can be successfully applied in these types of services, from the simplest, with universal applicability, to the most complex and specific. According to the functionality of the CI-focused methodologies, most of the tools used are related to the identification of the causes of ongoing problems, followed by planning, operations, and logistics tools, and common waste elimination tools. In specific, the DMAIC cycle is the most implemented tool. In fact, there is a growing trend towards the implementation of six sigma methodologies in different sectors of activity, and most often these methodologies follow the lean thinking concept. Research conducted in health care services and the manufacturing and agrifood industries are good examples of this (Costa et al., 2018; Samanta et al., 2023; Skalli et al., 2023; Thakur et al., 2023). Our results point towards a similar trend in the FS industry.

It is important to note that most of the studies reviewed use several different tools whose complementarity provides better results and synergistic improvements.

Regarding the outcomes, according to our analysis, operational and performance improvements, as well as customer satisfaction, are the most common results of implementing CI initiatives in the FS industry. The results obtained are mostly in line with the motivations/goals proposed by other authors and are transversal to the different FS sectors. On the other hand, there are still few studies in this context that seek to combine CI to obtain improvements in environmental sustainability practices. Only four of the reviewed studies sought to combine lean practices with sustainability. In three of them, significant quantitative results are presented: the first, a multiple case study in catering services, achieved a significant reduction of food waste in 5 out of 19 organisations (Strotmann et al., 2017); the second study, developed in a hospital, managed to reduce meal waste by about 50 % (Teeling et al., 2019); and the third study, carried out in a fast-food restaurant, achieved an improvement in energy efficiency, representing an important contribution to the mitigation of CO₂ emissions (Indrawati et al., 2020). There is a growing concern in the scientific community about environmental issues, namely in terms of responding to the SDGs proposed by the United Nations. CI is no exception, with a growing number of articles that integrate green practices with lean tools in several areas of activity (Marco-Ferreira et al., 2020).

In the present review, we also studied the factors that promote the successful implementation of CI. Most studies mention the significance of factors related to human issues in the context of the FS industry, with a particular emphasis on training and skills. Additionally, employee involvement and empowerment can counteract resistance to change and increase motivation and performance, driving CI to synergistic levels. We also found similar results in the food industry (Costa et al., 2018). Furthermore, a recent review of success factors and barriers in different sectors of activity identified top management involvement, project management and organisational infrastructure, and training and education as the main factors for success (Francescato et al., 2023).

This review also concluded that the complex organisational specificities of FSs are factors with a great impact on the strategic choice of CI-focused methodologies. Other research works on CI in the food industry also refer the complex organisational specificities of FSs as important barriers since they hinder streamlined production and inventory planning (Dora et al., 2016; Costa et al., 2018).

It should be emphasised that, as the classic works on CI state, the application of the kaizen principles described in the literature is fundamental to the successful implementation of CI in FS management. In this review, most of the principles were identified directly or indirectly in the success factors of the studies analysed, the most prominent

being the "people-oriented approach" guiding principle, demonstrated by the highest number of citations referring to human issues factors.

After identifying the factors of success or failure, it is interesting to see that there is a complex cause-and-effect relationship between the different types of factors mentioned which must be taken into consideration during the implementation of CI-focused methodologies.

Moreover, as described in the literature, several studies included in this review use different concepts interchangeably. In some cases, the methodology used is not identified, and in others their methodology is framed as *kaizen*, TQM, lean or six sigma, they end up implementing different tools and techniques indiscriminately, without really matching them to the methodology. In the restaurant and fast-food sectors, lean and six sigma were the most cited methodologies, and both VOC and VSM tools were used, while the DMAIC cycle was only used in the six sigma methodology. Although not treated as such in this review, in two included studies performed in the restaurant and fast-food sectors, *kaizen* appears as a methodology using the 5S and standardisation tools, and Ishikawa diagram and PDCA cycle, respectively. In the catering sector, a TQM implementation was carried out using the PDCA cycle and VSM tools. In the cafeterias, for example, it is worth highlighting the implementation of the lean methodology using the *spaghetti* diagram, 5S and standardisation tools. In the hospital facilities, 5S tool are also used combined to *kaizen blitz* are used as part of the lean methodology. On the other hand, two other studies carried out in the same sector used the lean six sigma methodology, where the DMAIC cycle, PDCA cycle and VSM tools were applied together in the first, and the DMAIC cycle and Ishikawa diagram tools in the second. At the university facilities, the most implemented methodology was six sigma and although DMAIC cycle stands out as a tool, other tools were also applied such as VOC and the Ishikawa diagram. Although this mishmash of tools and techniques ends up bringing positive results to organizations, as concluded in this review, it is important to conduct more studies in order to clarify the real results, success factors and limitations for each specific methodology or tool.

Regarding the limitations of this study, we can point out that due to the reduced number of articles available on the present topic, some caution should be exercised in generalising the conclusions. Additionally, some of the studies included in this review address the application of CI in FSs from different sectors, presenting the results together without distinguishing the improvements in each studied FS. Additionally, some other studies do not present quantitative and concrete results, making it difficult to perceive the degree of improvement promoted by the applied method.

6. Conclusions and future directions

To the best of our knowledge, this is the first article to systematically search and review the available scientific literature on continuous improvement in food service management.

The results obtained in this review make it possible to answer the research questions initially posed.

The methodologies most commonly mentioned in the literature dedicated to the food service industry relate to *problem identification and resolution*, through the Ishikawa diagram and Pareto chart tools, and *planning, operations, and logistics*, through the PDCA cycle and *kaizen* tools. The best results are obtained through the combination of different tools. Regarding the use of different tools, the results obtained are transversal to all the food service sectors studied. However, the limited number of studies found in certain sectors does not allow for an analysis of the frequency of application of the different tools for a particular type of food service. In addition, the present study shows that the main improvements achieved by implementing continuous improvement in the food service industry happen at the operational and performance level (87 %), followed by improvements in customer satisfaction (59 %). Finally, human issues are mentioned the most when it comes to the factors determining the success or failure of continuous improvement

implementations.

The present review is a significant contribution to food service management research and the information gathered could serve as a starting point for food services wishing to implement continuous improvement. To increase the scope of the review, we included and analysed studies addressing the diagnostic phase of continuous improvement, as well as studies completing the implementation of continuous improvement-focused tools. Additionally, we comprehensively examined the methodologies used in these studies and the benefits they promote in food service management. Furthermore, we gathered considerable information regarding the critical success factors, barriers, and limitations pertaining to the implementation of the described continuous improvement-focused methodologies. Researchers and managers who intend to study or implement projects in this area might find this information useful in preparing for possible challenges. Thus, we consider that the present review adds value to the existing literature, as it allows for a systematised, detailed and comparative analysis between the outcomes described in the literature.

Future studies should contribute with more research to cement the effectiveness of continuous improvement in the food service industry. It is necessary to understand how the performance of food services is maintained over time, as none of the studies reviewed included a long-term evaluation of the impact of continuous improvement-focused methodologies.

Although the results show that continuous improvement promote improvements in food services, the relationship between certain tools and the specific types of improvements achieved is still unclear. It could be useful to study the development of an ideal continuous improvement roadmap that meets the specific management needs of the food service sector and achieves effective quality improvements. There is also a need for more studies focusing on ways to counteract, mitigate, or eliminate barriers to successful continuous improvement implementations, particularly in the food service industry, which has peculiar and unique characteristics related to the diversity of ingredients and meals produced, as well as specific and mandatory requirements that represent barriers to continuous improvement. Furthermore, more studies should be carried out in hospital and long-term care settings, as the literature dealing with this type of food service is scarce. In terms of the institutional sector, it could be useful to conduct more research on particularities such as target customers and stakeholders, as well as organisational and economic conditions, which can be a challenge to the implementation of continuous improvement.

And lastly, it would be important to conduct more research on the benefits of sustainability-oriented continuous improvement, particularly to environmental sustainability, as this could make sustainable solutions more affordable for the food service industry.

CRedit authorship contribution statement

Vera Miguéis: Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Cláudia Afonso:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Célia Monteiro:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Ada Rocha:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of Competing Interest

none.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijhm.2024.103916.

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